



ANALYZING THEMATIC ALIGNMENT IN DRUG SAFETY

Do published articles really match what a journal says it publishes?

NLP - Exam Project
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- EVERY JOURNAL DECLARES AN **AIMS & SCOPE**
- EDITORIAL FOCUS CAN **DRIFT OVER TIME**
- READING EVERYTHING BY HAND DOESN'T SCALE

RESEARCH QUESTION

To what extent are the articles published in *Drug Safety* thematically aligned with the journal's stated Aims & Scope, and how has this alignment evolved between 2015 and 2025?

THE DATASET

DRUG SAFETY

Scientific journal

PUBMED

Domain

1,074

Valid articles

2015–2025

Year range

EDA & PREPROCESSING:

- **Year 2026**
- **“Duplicate” title**
- **Correction notices**
- **Text preprocessing**



Only 39 articles vs. 89–115 for a complete year → **excluded** as incomplete

Investigated: two distinct correction notices, **not a real duplicate**

16 short **“Correction to:”** notices **removed** (not real research abstracts)

Whitespace **normalized**; wording and casing **preserved**

TURNING TEXT INTO VECTORS

01 CHUNKING

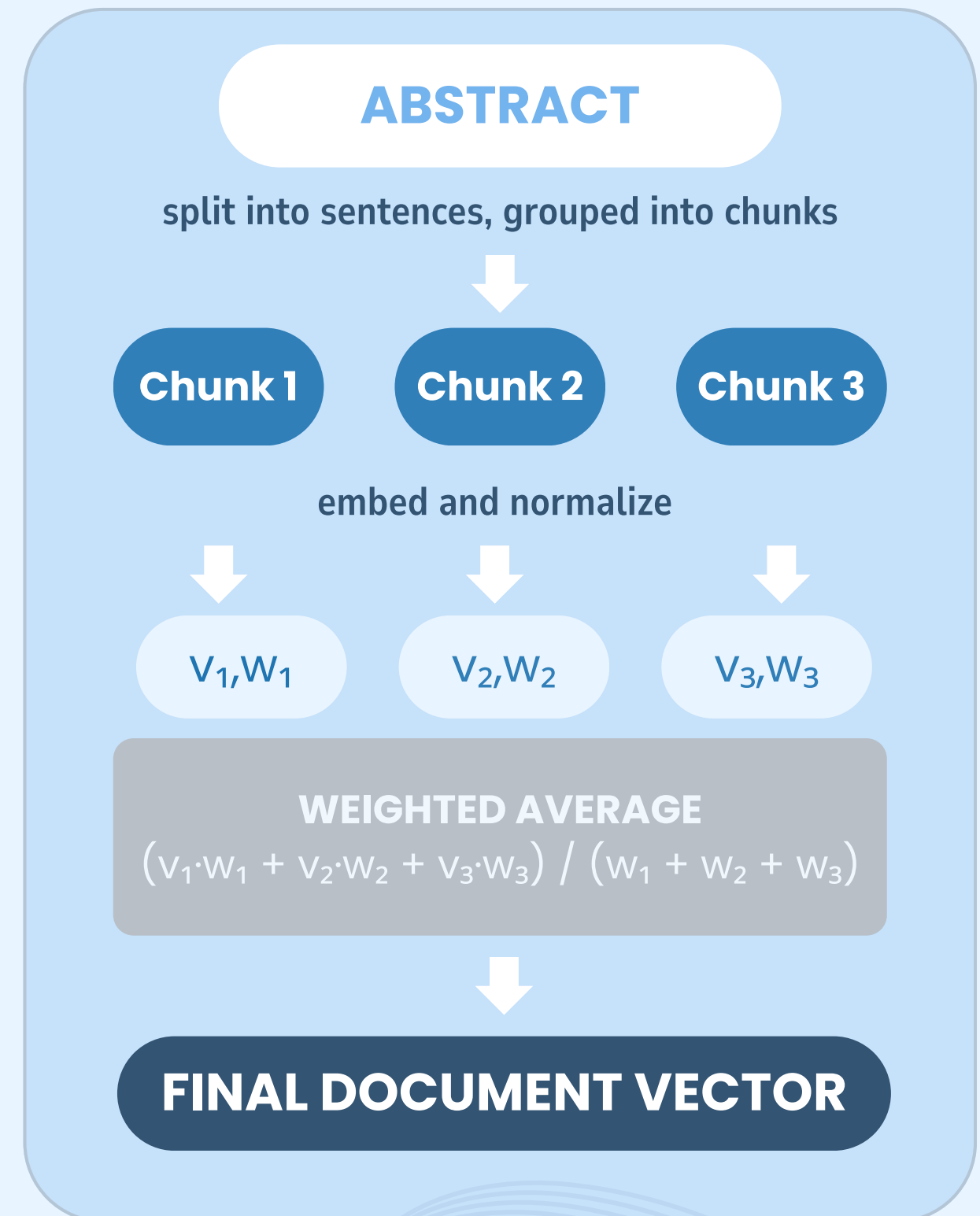
The problem: **abstracts are too long**

92.3% of the final corpus needed more than 1 chunk

Our solution: **sentence-based chunking**

02 EMBEDDING

SentenceTransformer: all-MiniLM-L6-v2



w_i = number of tokens in chunk i

ALIGNMENT SCORE AND OUTLIERS

Alignment score

$$\text{Alignment}(i) = \cos(v_i, v_s) = \frac{v_i \cdot v_s}{\|v_i\| \|v_s\|}$$

cosine similarity between each abstract's vector and the Aims & Scope vector

Outlier detection: z-score, threshold 2.0

$$z_i = \left| \frac{\text{Alignment}(i) - \mu}{\sigma} \right| > 2.0$$

49 outliers found:

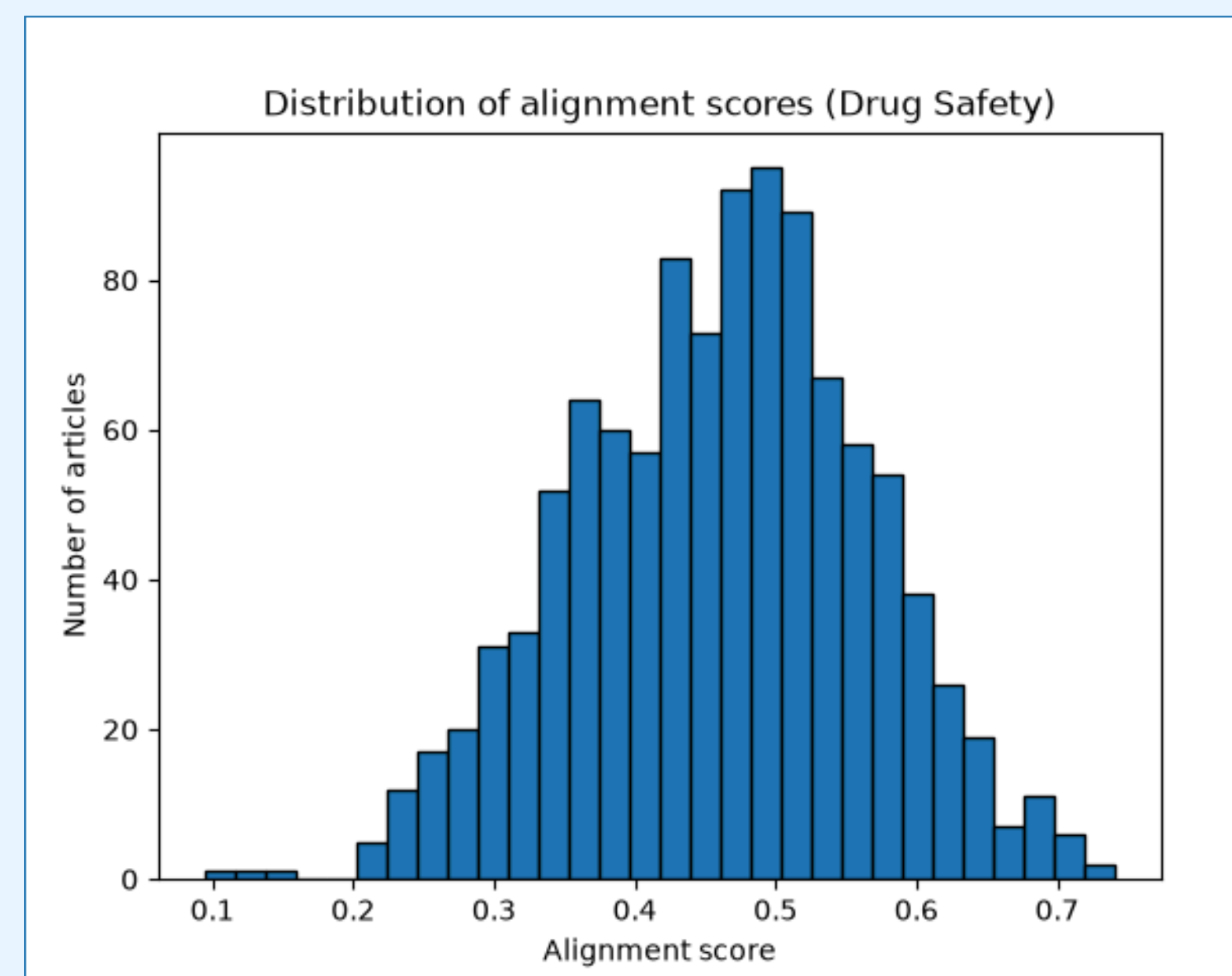
- Low-alignment outliers: 28
- High-alignment outliers: 21

0.461

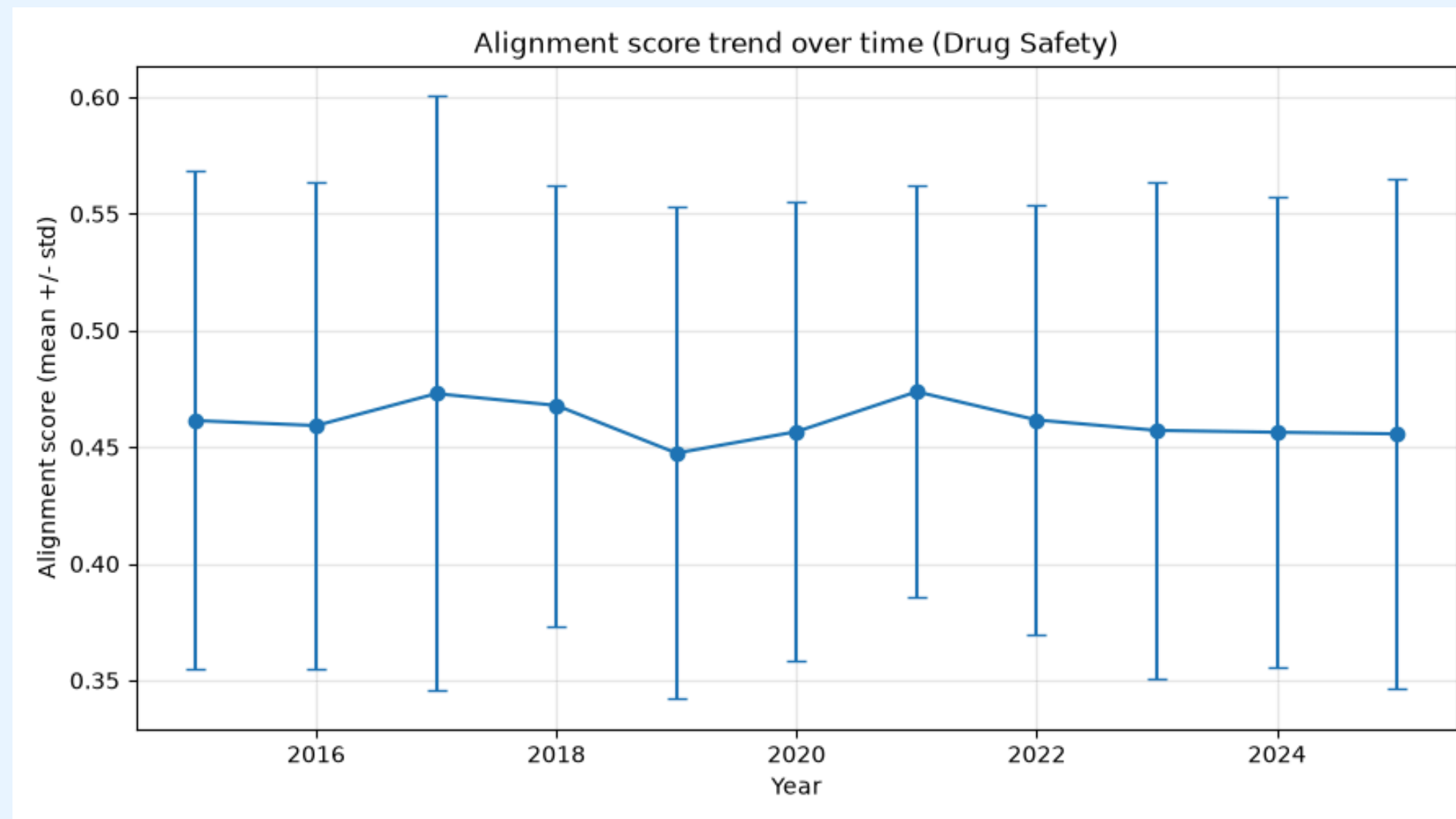
Mean score

0.104

Std. dev.



ALIGNMENT OVER TIME



A stable picture, on average

No systematic drift in the yearly average: stays within 0.45–0.47 across the whole decade.

But does that mean nothing changed underneath?

MODEL CHOICE VALIDATION

Why Sentence-BERT and not raw BERT?

Sentence-BERT provides more discriminative similarity scores

Sentence-BERT

Mean 0.461

SD 0.100

Semantic-similarity optimized

Naive BERT

Mean 0.841

SD 0.025

Raw mean pooling

Correlation: $r=0.339$, $p<0.001$

Naive BERT compresses scores into a narrow high-similarity range,
while Sentence-BERT better separates articles by semantic alignment.

READING THE EXTREMES

HIGHEST ALIGNMENT

0.741

Navigating Diverging Perspectives: Reasoning, Evidence, and Decision-Making in Drug Safety

0.727

Summarising the Evidence for Drug Safety: Meta-Analysis Approaches

0.716

READUS-PV: Explanation and Elaboration

All pharmacovigilance methodology: decision frameworks, meta-analysis, reporting standards.

LOWEST ALIGNMENT

0.094

LLM Performance & Reproducibility in Named Entity Recognition

0.124

Adverse Effects After HPV Vaccination — Japan

0.158

Narrative Search Engine with AI Query Suggestions

Two on AI/ML methods, one on vaccine safety with narrow, specific vocabulary.

BERTopic

From individual articles to thematic patterns

BERTopic identified 24 topics + 1 noise cluster

HIGHEST ALIGNMENT

Topic 8

Pharmacovigilance
0.585

Navigating Diverging Perspectives:
Reasoning, Evidence, and Decision-
Making in Drug Safety

Topic 16

Disproportionality
0.566

Summarising the Evidence for
Drug Safety: Meta-Analysis
Approaches

Topic -1

Noise
0.491

READUS-PV: Explanation and
Elaboration

LOWEST ALIGNMENT

LLM Performance &
Reproducibility in Named Entity
Recognition

Adverse Effects After HPV
Vaccination — Japan

Narrative Search Engine with AI
Query Suggestions

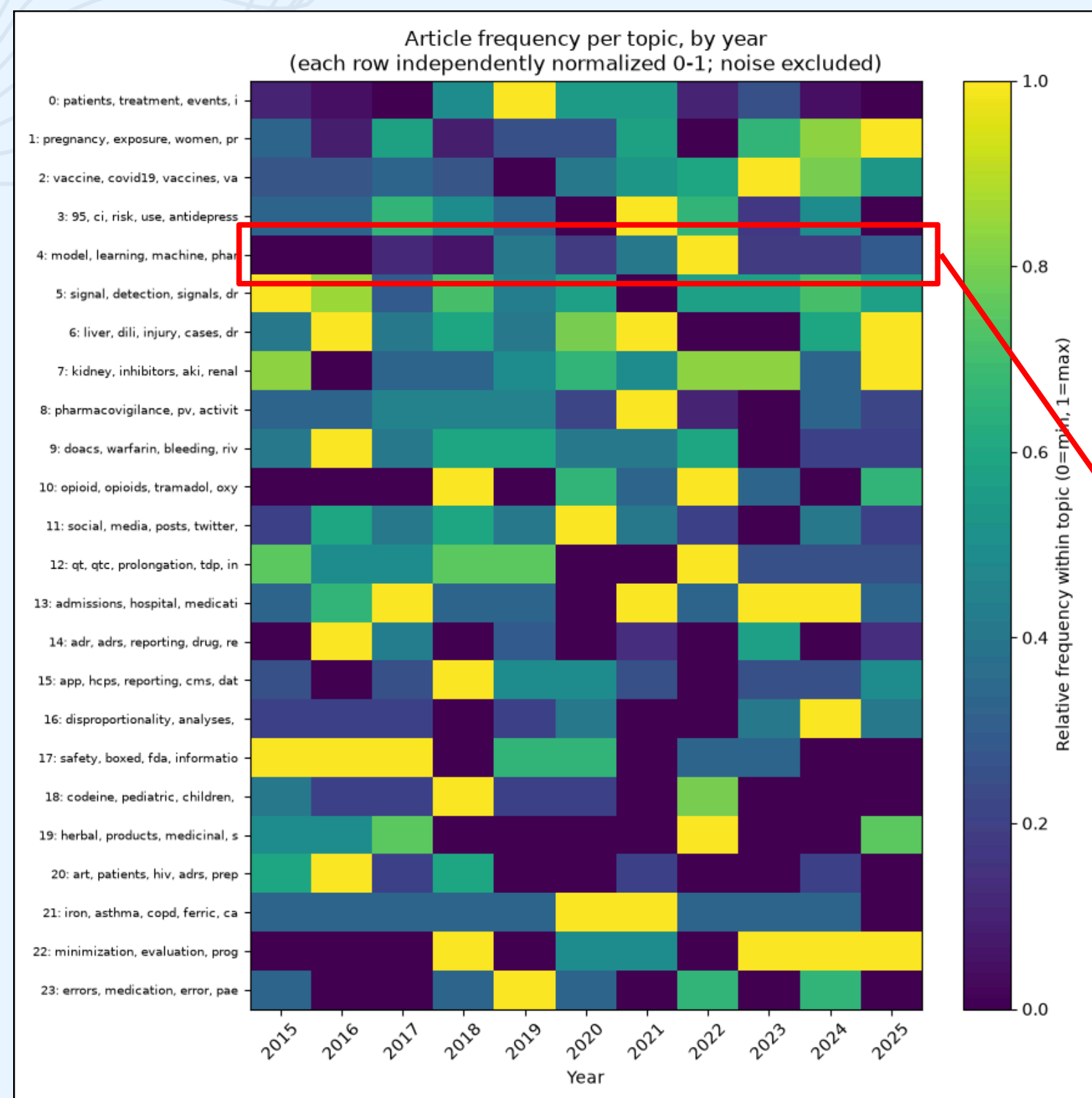
Topic 4

AI / Machine Learning
0.409

Topic 2

Vaccines
0.344

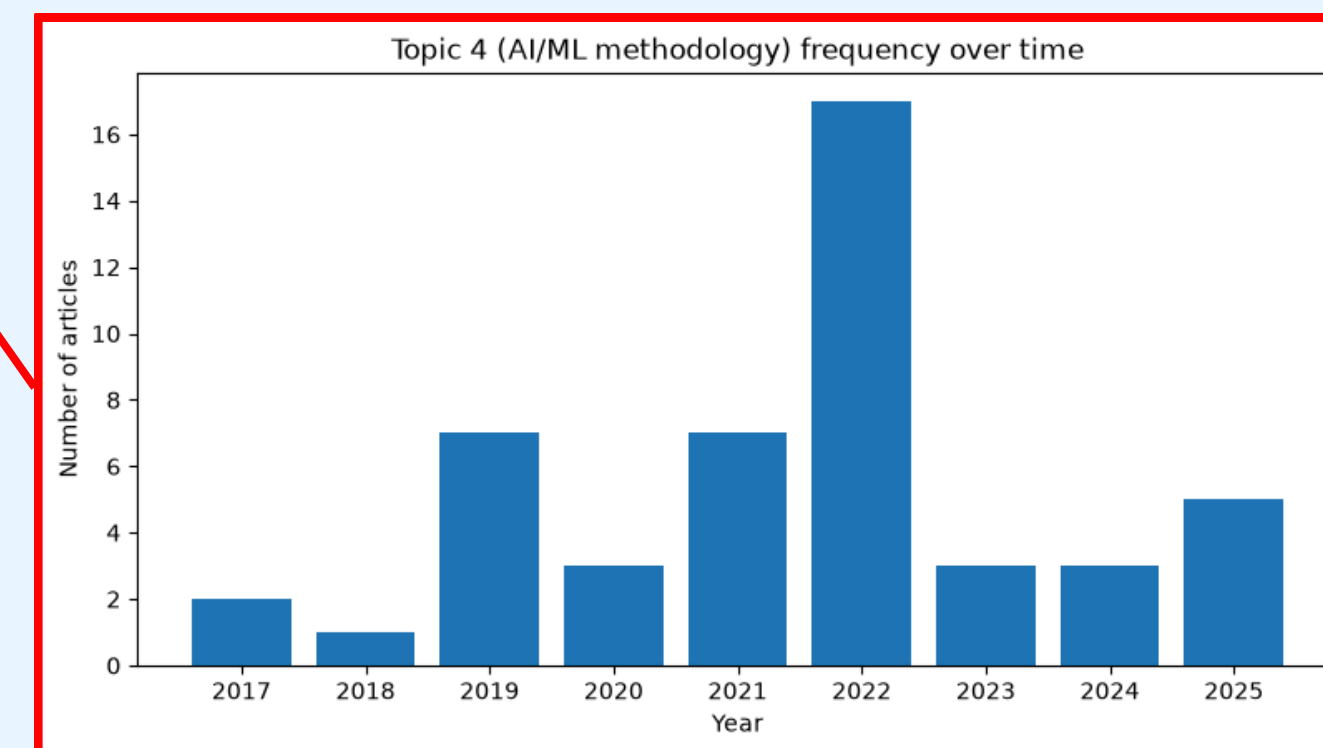
WHAT THE FLAT TREND WAS HIDING



Stable alignment

$\neq$

Stable thematic composition



EXAMPLE: Topic 4 — AI / ML

CONCLUSIONS

01

Overall alignment is moderate and **stable over time**

02

Pharmacovigilance-related topics show the strongest alignment

03

Thematic composition **changes** despite a stable aggregate score



THANK YOU